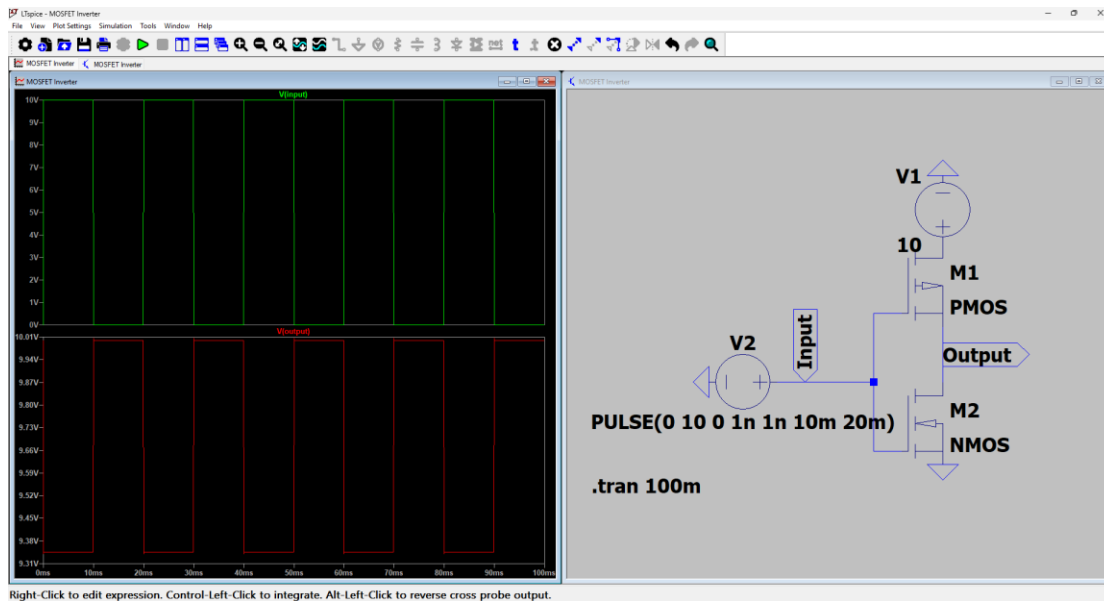


MOSFET Logic Gates

ENGS152 Circuits Lab

18 April 2026

Step 1: Inverter



To understand how the inverter works we have to understand how each MOSFET behaves under certain conditions. PMOS turns ON as the gate voltage is LOW and becomes OFF when the gate is HIGH. NMOS behaves the opposite way it turns ON as the gate is HIGH and turns off as the gate becomes LOW.

INVERTER MOSFET PRINCIPLE

Input: LOW (0V) → PMOS turns ON → HIGH (10V) passes to the output node.

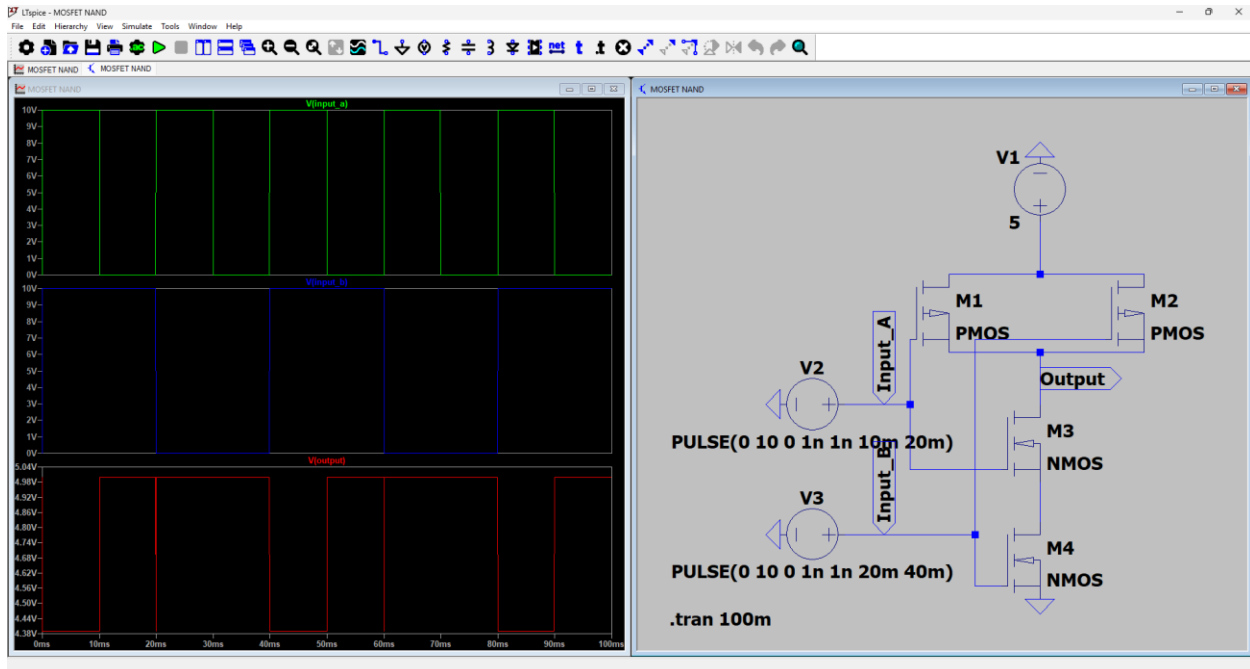
NMOS turns OFF → HIGH (10V) has no way to the ground so it passes through output.

Input: HIGH (10V) → PMOS turns OFF → HIGH (10V) has no road to the output node.

NMOS turns ON → Connecting the output node to the GND.

Like this the output becomes the inverse of the input.

Step 2: NAND Gate



NAND GATE MOSFET PRINCIPLE

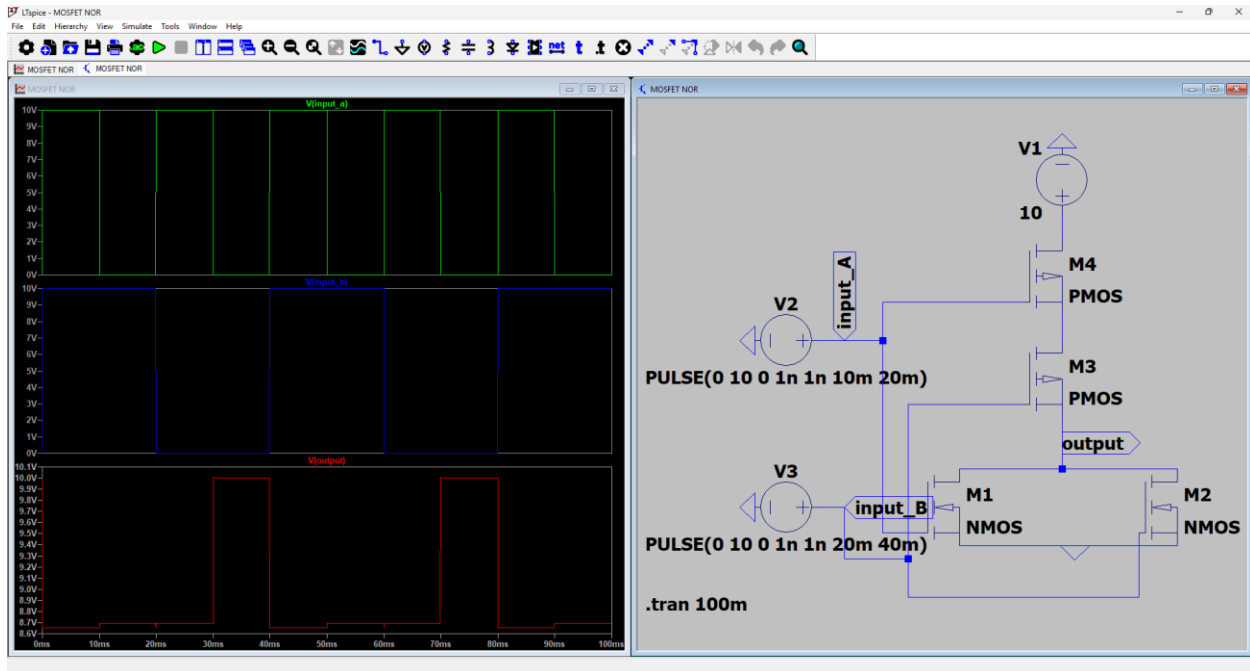
PMOS are connected in parallel, while NMOS are connected in series.

A	B	PMOS A	PMOS B	NMOS A	NMOS B	X
0	0	1	1	0	0	1
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	1	1	0

** Please note, in this table 1 means the transistor is ON, while 0 means the transistor is OFF. **

As we can see, whenever any of the signals is 0, one of the PMOS transistors turns ON and at least one of the NMOS transistors turns OFF, and the output receives the supply's voltage and at the same time disconnects from the GND. However, when both signals are 1, both PMOS turn OFF, both NMOS turn ON, and the output gets connected to GND, receiving value of 0.

Step 3: NOR Gate



NAND GATE MOSFET PRINCIPLE

PMOS are connected in series, while NMOS are connected in parallel.

A	B	PMOS A	PMOS B	NMOS A	NMOS B	X
0	0	1	1	0	0	1
0	1	1	0	0	1	0
1	0	0	1	1	0	0
1	1	0	0	1	1	0

** Please note, in this table 1 means the transistor is ON, while 0 means the transistor is OFF. **

As we can see, whenever at least one of the signals is 1, at least one of the PMOS transistors turns OFF, it breaks the circuit from supply to output, and at least one NMOS transistor becomes ON, and the output doesn't receive the supply's HIGH voltage, but obtains value from the GND. However, when both signals are 0, both PMOS turn ON, both NMOS turn OFF, and the output gets connected supply, obtains the 1 value, and at the same time the circuits breaks from output to GND, forcing the supply voltage to pass only through the output node.